Vertically Integrated Projects

Catalogue description

In VIPP courses, students engage in applied research with faculty and graduate students on multi-disciplinary, team-based research projects. Students work with faculty and industry mentors on real world large-scale projects with students from different majors. Participants develop **research experience** and **team skills**, while integrating knowledge from previous courses and refining their professional skills (teamwork, project planning, and communications).

About VIP

A Vertically-Integrated Projects (VIP) Program operates in a research and development (R&D) context. Undergraduate students who join VIP teams earn academic credit for their participation in design/discovery efforts that assist faculty and graduate students with research and development issues in their areas of expertise.

The teams are:

- *Multidisciplinary* - drawing students from all disciplines on campus;
- *Vertically-integrated* - maintaining a mix of undergraduate and graduate students each term;
- *Long-term* – while the project is long term and large scale, students get to work on sub-themes.

The continuity, technical depth, and disciplinary breadth of these teams are intended to:

- Provide the time and context necessary for students to **learn and practice** many different professional skills, make substantial **contributions** to the project, and experience many **different roles** on a **large, multidisciplinary** VIP team.
- Support long-term interaction between the graduate and undergraduate students on the team. The graduate students mentor the undergraduates as they work on VIP projects embedded in the graduate students' research.
- Enable the completion of **large-scale projects** that are of significant benefit to faculty members' research programs.

Learning outcomes

1. Learn and practice professional skills;
2. Make substantial contributions to a team project;
3. Experience different roles on a large, multidisciplinary team;
4. Investigate scientific questions as part of a well-organized research project;
5. Present and defend research findings among peers;
6. Be better prepared for undertaking graduate work.

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Timeline

Week	Activity/Event
Week 1	Introductions, overview of team's work, discussion of semester goals
Week 2	Sub-team selections finalized, and sub-team meeting times finalized
Week 3	Verification of student participation in class.
Week 8	Midterm presentations
Week 12	Web-based peer-evaluations released for students to complete. Late submissions will not be accepted
Week 14	Final presentations. Turn in VIP notebooks at team meeting for final

Student evaluation

The premise of VIP is teams working on projects. Much like a real-world team, individual members work on different aspects of the project. The grade is based on three areas. Students must demonstrate achievements in all three areas below.

1. Documentation and records (33%)
    a. VIP Notebook
    b. Saving all of your work on our Teams Space
    c. Code (via GT GitHub)
    d. Technical report
    e. Demo video and/or live walkthrough
    f. Poster for final showcase
2. Personal accomplishments and contributions to the team's goals (33%)
    a. Quizzes, learning modules, essays, reports required by the advisor(s);
    b. Engagement in project;
    c. Pursuit of knowledge necessary for project;
    d. Contributions to the technical progress of the team;
3. Teamwork and interaction (33%)
    a. Peer Evaluations;
    b. On-time attendance in meetings;
    c. Actively contributes to overall team goals;
    d. Coordinates activities with other team members;
    e. Assists other team members;
    f. Team presentation(s).

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Overview: Context-Aware, Learning-Driven Autonomous Ground Vehicles for Developing Countries

This guide organizes the full set of autonomous vehicle (AV) projects across three tiers of increasing complexity. Each project includes:

- **Objectives**
- **Key Deliverables**
- **Simulation Environment**
- **Algorithms / Tools**
- **Assessment Criteria**
- **Possible Extensions**
- **Student-Facing Assignment Sheet**

Recommended Tools Across Tiers

Tool	Use Case
Python + Gym	Basic simulation & RL
PyTorch / TensorFlow	RL agent training
CARLA / Gazebo	High-fidelity AV simulation
SUMO / OSMnx	Traffic and map-based routing
ROS2	Full system integration
Jetson Nano / Raspberry Pi	Real-world AGV deployment

TIER 1 – Foundational Projects

These projects aim to build a **foundation in reinforcement learning, basic multi-agent interactions, and rule-based decision-making**.

❖ Project 1: Physics-Aware Local Path Following using Basic RL

Objectives:

- Train an RL agent to follow a predefined path while respecting basic vehicle kinematics and avoiding static obstacles.

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Key Deliverables:

- A working simulation of an AV following a path.
- Visual outputs (trajectory plots, reward curves).
- Report analyzing learned policy and performance.

Simulation Environment:

- Python + PyBullet, Box2D, or simple 2D simulator.

Algorithms / Tools:

- Q-Learning / DQN
- Python, NumPy, PyTorch/TensorFlow

Assessment Criteria:

- Collision avoidance
- Path accuracy
- Compliance with kinematics

Possible Extensions:

- Introduce moving targets or dynamic obstacles.

❖ Project 2: Basic Multi-Agent Simulation for Aggressive Traffic Interaction

Objectives:

- Simulate an AV interacting with aggressive drivers and learn avoidance strategies.

Key Deliverables:

- Multi-agent simulation
- MARL policy implementation
- Behavior visualizations

Simulation Environment:

- SUMO, PettingZoo, or custom gridworld

Algorithms / Tools:

- Independent Q-Learning / Shared Policy MARL
- Discrete actions (acceleration, turning)

Assessment Criteria:

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- Collision frequency
- Time to goal
- Behavior under pressure

Possible Extensions:

- Introduce partial observability or communication.

❖ Project 3: High-Level Decision Learning for Informal Intersections

Objectives:

- Learn decision policies for navigating unstructured intersections with minimal signage.

Key Deliverables:

- 2D simulation of intersection
- Decision-making policy (RL or supervised)
- Report on safety and efficiency

Simulation Environment:

- Custom 2D sim with pygame, Unity ML-Agents, or simple top-down simulator

Algorithms / Tools:

- Rule-based + tabular RL or decision trees

Assessment Criteria:

- Collision rate
- Delay/time-to-cross
- Policy interpretability

Possible Extensions:

- Add pedestrian crossings or variable driver behavior

TIER 2 – Intermediate Complexity

These projects introduce terrain-awareness, more complex multi-agent systems, and uncertainty.

❖ Project 4: Terrain-Adaptive Navigation Using RL

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Objectives:

- Train an RL agent that adapts its path in response to terrain (e.g., potholes, gravel, uneven surfaces).

Key Deliverables:

- Terrain-aware planner
- Simulation with varying terrain types
- Report with terrain-avoidance metrics

Simulation Environment:

- CARLA, Gazebo, or 2D grid terrain cost map

Algorithms / Tools:

- PPO, SAC
- Reward: penalize bumpy or high-cost terrain

Assessment Criteria:

- Smoothness
- Terrain avoidance
- Policy robustness

Possible Extensions:

- Add weather effects (e.g., slippery roads)

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❖ Project 5: Multi-Agent Traffic Behavior Modeling with Aggressive and Defensive Drivers

Objectives:

- Simulate AV interactions with both aggressive and defensive agents to develop adaptive response strategies.

Key Deliverables:

- Multi-agent simulation
- Behavior policies for human-like agents
- Evaluation of AV strategy

Simulation Environment:

- CARLA with scripted behaviors or SUMO

Algorithms / Tools:

- RL + rule-based imitation
- Observation: speed, proximity, intent

Assessment Criteria:

- Safety performance
- Agent negotiation ability
- AV adaptability

Possible Extensions:

- Social-awareness or communication between agents

❖ Project 6: High-Level Decision-Making for Mixed Traffic and Informal Intersections

Objectives:

- Equip AV with high-level decision policy to navigate informal intersections with mixed traffic (cars, pedestrians).

Key Deliverables:

- Mixed-agent simulation
- High-level policy implementation
- Visual and metric evaluation

Simulation Environment:

- CARLA / SUMO with human-pedestrian agents

Algorithms / Tools:

- Hierarchical RL, decision trees
- Pedestrian modeling

Assessment Criteria:

- Decision latency
- Safety compliance
- Collision and wait time

Possible Extensions:

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- Ethical decision-making layer

TIER 3 – Advanced Integration and Real-World Testing

These capstone-level projects combine perception, planning, control, and real-world deployment.

❖ Project 7: Physics-Aware Local Planning with Advanced RL for Safety

Objectives:

- Develop a real-time, physics-aware local planner using advanced RL and vehicle models.

Key Deliverables:

- RL-based planner
- Dynamic obstacle avoidance
- Policy report with safety metrics

Simulation Environment:

- CARLA or AirSim with full physics

Algorithms / Tools:

- SAC / DDPG + vehicle model (bicycle or dynamic)
- Reward includes smoothness, risk zones, speed limits

Assessment Criteria:

- Real-time performance
- Constraint satisfaction
- Smoothness and stability

Possible Extensions:

- Add probabilistic safety margins

❖ Project 8: Full-Stack Integration and Testing of Autonomous Vehicles

Objectives:

- Build and integrate a full AV stack and validate performance in simulation or hardware.

Key Deliverables:

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- Perception → Planning → Control pipeline
- ROS integration
- Simulated or real-world tests

Simulation Environment:

- CARLA or Gazebo; optional JetRacer/TurtleBot hardware

Algorithms / Tools:

- ROS2, YOLO/SSD for detection
- MPC / Pure Pursuit for control

Assessment Criteria:

- Integration quality
- Latency, safety, functionality
- Documentation and demo quality

Possible Extensions:

- Sensor fault handling or low-light testing

❖ Project 9: Robust Multi-Agent Cooperative Planning for Complex Urban Scenarios

Objectives:

- Develop a multi-agent cooperative planning system where multiple autonomous vehicles coordinate maneuvers in dense urban traffic with mixed human and autonomous agents.
- Focus on communication protocols, safety guarantees, and efficiency in complex traffic (e.g., merging, lane changes, roundabouts).

Key Deliverables:

- Multi-agent simulation with AVs coordinating actions using cooperative RL or game-theoretic planning.
- Communication protocol implementation (e.g., message passing, shared intentions).
- Safety and efficiency analysis comparing cooperative vs. independent strategies.
- Visualization of multi-vehicle trajectories and interactions.

Simulation Environment:

- CARLA or SUMO with multiple AV agents.

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- Possible integration with ROS for real-time multi-agent control.

Algorithms / Tools:

- Multi-Agent Reinforcement Learning (e.g., MADDPG, QMIX).
- Game-theoretic or decentralized planning algorithms.
- Communication protocols or message passing frameworks.

Assessment Criteria:

- Collision avoidance and safety margins.
- Traffic throughput and efficiency metrics.
- Scalability of coordination with increasing agent numbers.
- Robustness to communication delays or failures.

Possible Extensions:

- Incorporate uncertainty in agent intentions.
- Extend to heterogeneous agent types (cars, pedestrians, cyclists).
- Implement fallback decentralized planning for communication loss.

Other project ideas

❖ Adaptive Pedestrian Detection and Crossing Behavior for AVs

Description: Train an AV to safely interact with pedestrians by developing a robust pedestrian detection system and a decision-making model for predicting pedestrian crossing behavior. This would also explore the ethical considerations of how an AV should react to pedestrians in unpredictable scenarios.

- **Focus**: Computer vision, decision-making under uncertainty, safety, ethics in AV decision-making.

❖ Collision Avoidance in Dense Urban Environments using Real-Time Path Planning

Description: Develop an advanced collision avoidance algorithm using sensor data (e.g., LiDAR, cameras) to continuously adjust an AV's trajectory in dense traffic environments. The system should ensure safety in complex urban scenarios, such as intersections, pedestrians, and cyclists.

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- **Focus**: Real-time path planning, obstacle detection, safety constraints, dynamic environments.

❖ Adaptive Control for Autonomous Vehicles on Uneven and Slippery Roads

Description: Design an adaptive control system that ensures an AV maintains stability and safety on uneven or slippery roads. The system should be able to react to sudden changes in road conditions (e.g., ice patches, potholes) in real-time.

- **Focus**: Control systems, terrain adaptation, stability.
- **Tools**: MATLAB, Simulink, Python, vehicle dynamics models.

❖ Risk-Aware Autonomous Decision Making for Complex Urban Scenarios

Description: Develop a risk-aware decision-making system that allows the AV to weigh risks associated with various driving maneuvers (e.g., lane changes, turning at intersections) based on surrounding traffic, road conditions, and environmental factors.

- **Focus**: Risk assessment, decision theory, safety-critical systems.
- **Tools**: MATLAB, CARLA, Python.

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